

Echoes in the Model: When Features Reflect Predictions

*Trust-Aware Joint Feature-Prediction Discrepancy
for Robust Domain Adaptation*

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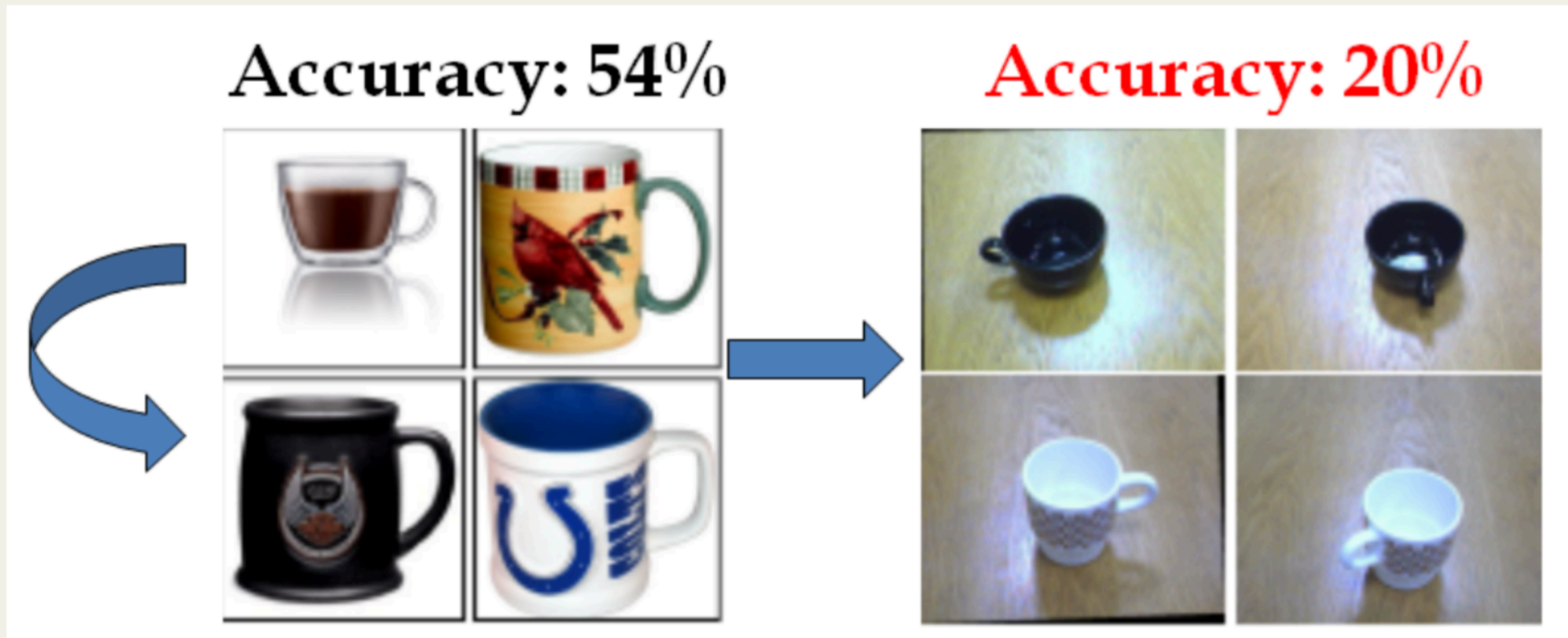
The Fragility of Machine Learning Models



Poor generalization ability

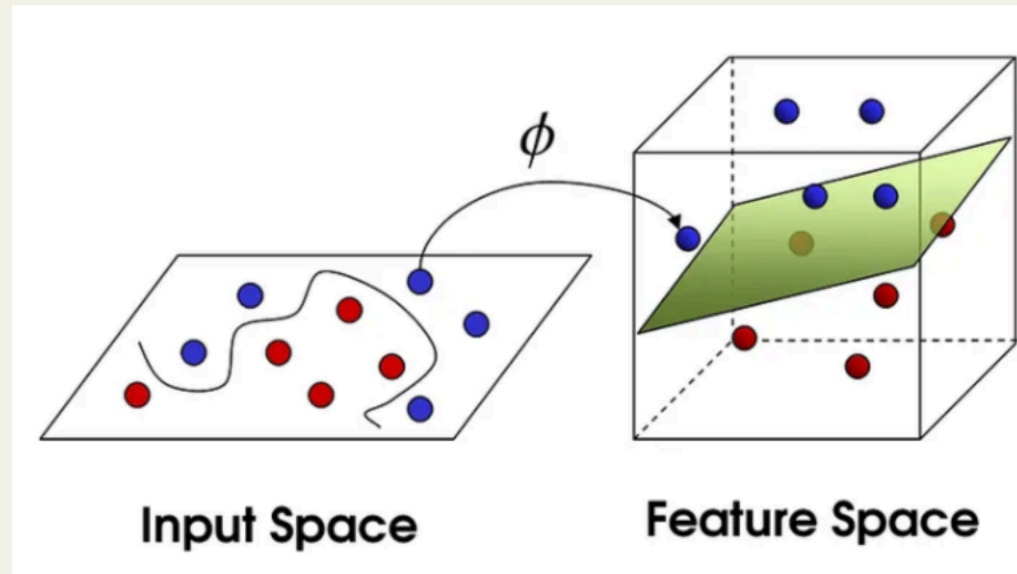


Credit: Tackling Domain Shift in AI: A Deep Dive into Domain Adaptation by Housseem Ben Salem



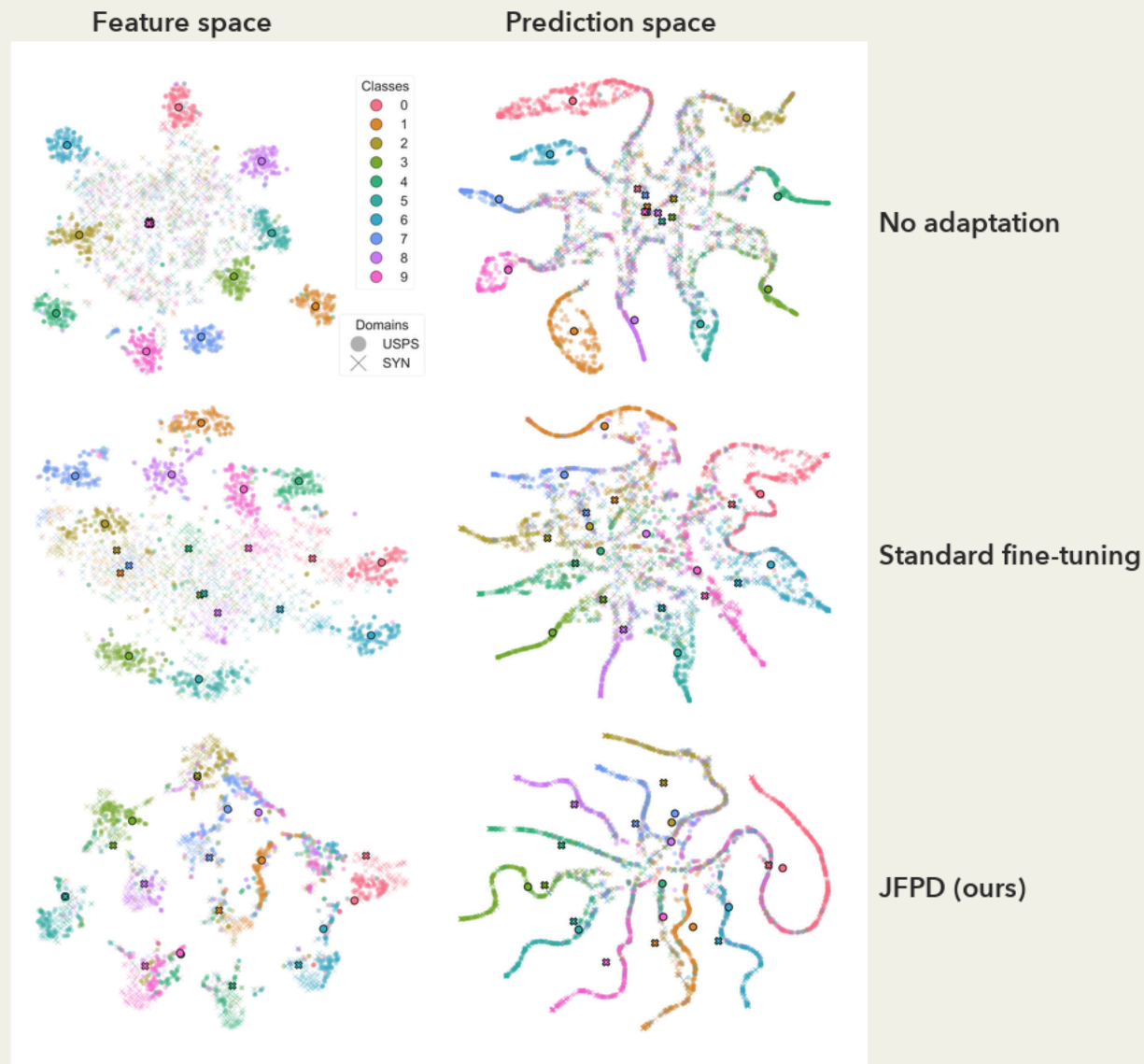
Credit: Domain Adaptation for Object recognition by Kate Saenko

Noisy and semantically ambiguous feature representations



Credit: Representations: The Most Important Thought Framework in Machine Learning

Disconnect between **features** and **predictions**



USPS (source, dots) → Syn (target, crosses)

- **Feature-based alignment** (MMD, cosine, Euclidean) ignores whether aligned features lead to consistent predictions.
- **Prediction-based alignment** (KL, JS) ignores the representation structure that produces those predictions.
- **Both treat every target sample as equally reliable**, so noisy or ambiguous samples mislead the alignment.

What if a model's features could echo its own predictions?



Trust-Aware Solution



Trust Modeling: Which Samples Can We Rely On?

Normalized divergences (bounded in $[0, 1)$)

$$d_{\text{feat}}(\mathbf{f}_s, \mathbf{f}_t) = \frac{d(\mathbf{f}_s, \mathbf{f}_t)}{1 + d(\mathbf{f}_s, \mathbf{f}_t)}.$$

$$d_{\text{pred}}(\mathbf{p}_s, \mathbf{p}_t) = \frac{D(\mathbf{p}_s, \mathbf{p}_t)}{1 + D(\mathbf{p}_s, \mathbf{p}_t)}.$$

d = cosine distance in feature space; D = JS divergence between softmax outputs

Uncertainty-aware trust ψ

$$\psi = \frac{1}{1 + H(\mathbf{p}_s) + H(\mathbf{p}_t)}.$$

Low prediction entropy in both domains $\rightarrow$ high trust; ambiguous predictions are down-weighted.

Semantic-alignment trust ϕ

$$\phi = \frac{1}{1 + d_{\text{feat}}(\mathbf{f}_s, \mathbf{f}_t)}.$$

Pairs already close in feature space $\rightarrow$ high trust; poorly aligned pairs are suppressed.

Cross-guided: prediction confidence ψ weights the feature term; feature alignment ϕ weights the prediction term.

Our Frame work: Joint Feature-Prediction Discrepancy (JFPD)

(i) a trust-weighted discrepancy metric

We combine divergence and trust into the unified JFPD metric:

$$d_{\text{JFPD}} = \alpha \cdot d_{\text{feat}} \cdot \psi + (1 - \alpha) \cdot d_{\text{pred}} \cdot \phi,$$

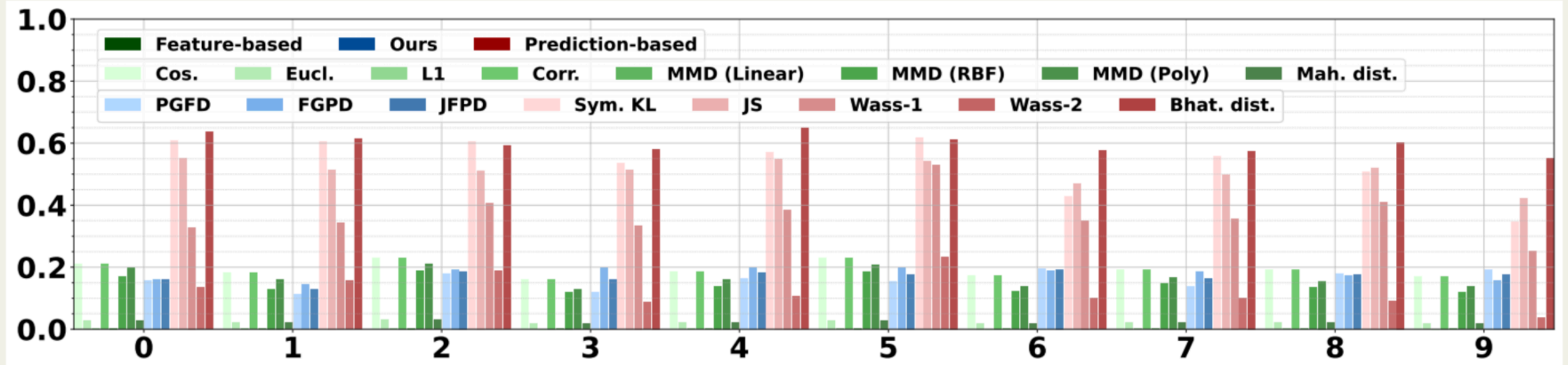
where $\alpha \in [0, 1]$ balances the semantic and categorical components.

(ii) a corresponding fine-tuning loss derived from this metric

$$\mathcal{L}_{\text{JFPD}} = \alpha \cdot \underbrace{(d_{\text{feat}}(\mathbf{f}_t, \mathbf{z}_{y^t}^s) \cdot \psi)}_{\text{Prediction-Guided Feature Discrepancy}} + (1 - \alpha) \cdot \underbrace{(d_{\text{pred}}(\mathbf{p}_t, \mathbf{p}_{y^t}^s) \cdot \phi)}_{\text{Feature-Guided Prediction Divergence}},$$

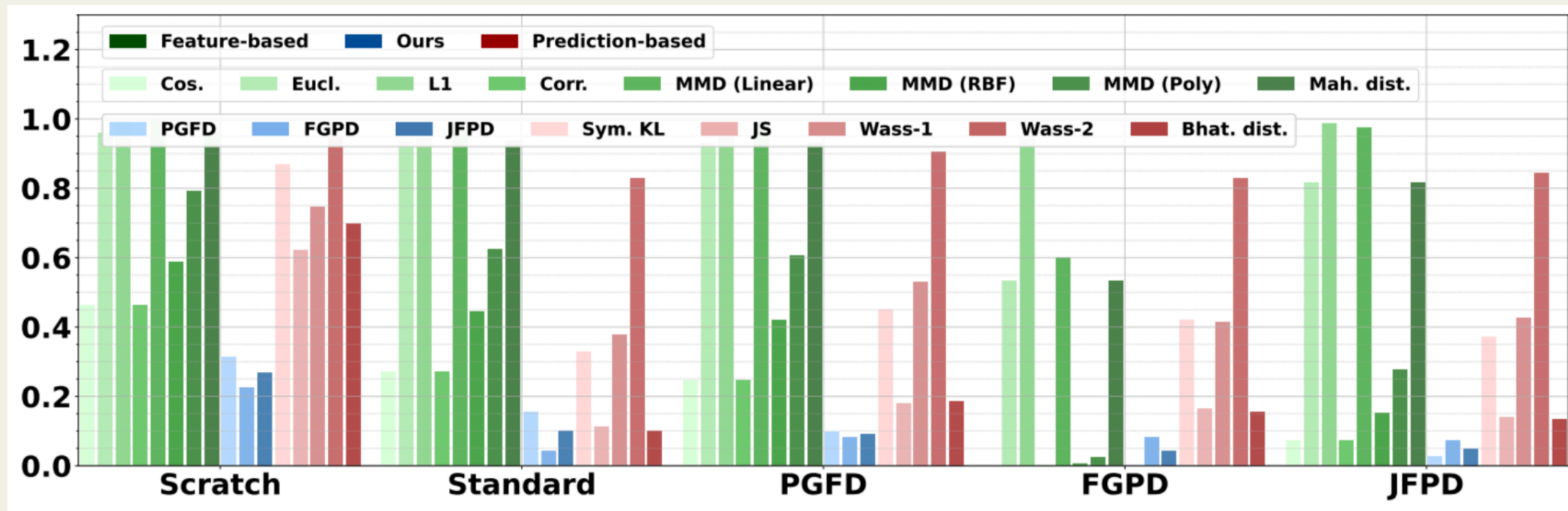
JFPD combines feature and prediction differences, weighted by these trust scores.

Class-wise Δ -divergence comparsion



The vertical axis shows the change in estimated domain gap per digit class and metric, computed as baseline (trained on MNIST only) minus fine-tuned (trained on MNIST, then adapted to SVHN).

Domain gap metrics across fine-tuning methods



Domain gap comparisons for models trained from scratch and fine-tuned using standard cross-entropy, our PGFD, FGPD, and JFPD. Bars represent various feature- and prediction-based discrepancy measures. JFPD consistently reduces domain gaps across both levels, most notably in feature-based metrics, highlighting the need for joint alignment beyond prediction scores.

Experiment results: Digits

TABLE I: Results on Digits. We evaluate two backbones: a ViT-S-4/9-420 (a small Vision Transformer with 4×4 patch size, 9 layers, and 420-dimensional embeddings) and a lightweight VGG-style CNN (three Conv $\rightarrow$ ReLU $\rightarrow$ Pool blocks).

Model	Source	Target															
		No fine-tuning				Standard fine-tuning				FGPD fine-tuning				PGFD fine-tuning			
		MNIST	SVHN	USPS	Syn	MNIST	SVHN	USPS	Syn	MNIST	SVHN	USPS	Syn	MNIST	SVHN	USPS	Syn
ViT-S	MNIST	99.65	19.88	94.02	18.80	–	94.06	97.46	75.35	–	95.01	98.16	88.50	–	76.64	97.76	60.40
	SVHN	70.34	93.79	76.53	39.35	99.49	–	97.01	80.15	99.71	–	97.91	87.95	95.63	–	95.62	52.60
	USPS	54.17	16.15	96.66	17.15	99.32	91.58	–	66.15	99.43	92.30	–	77.50	81.70	39.40	–	23.65
	Syn	20.33	30.95	62.33	73.30	99.29	91.92	96.66	–	99.47	93.80	97.36	–	69.96	35.60	78.03	–
CNN	MNIST	99.61	20.76	85.70	21.40	–	91.57	98.16	74.50	–	92.63	98.46	88.75	–	70.21	97.11	62.55
	SVHN	68.65	94.51	75.44	43.35	99.37	–	97.36	81.90	99.50	–	97.91	91.25	93.67	–	90.88	57.00
	USPS	94.31	20.62	97.66	19.60	99.46	91.44	–	70.50	99.56	91.78	–	83.65	98.28	47.10	–	39.85
	Syn	28.82	38.59	63.08	94.85	99.28	91.13	96.76	–	99.44	91.40	97.76	–	85.99	63.14	84.16	–

- Four domains (MNIST, SVHN, USPS, Synthetic Digits); each serves once as source $\rightarrow$ 12 transfer tasks.
- Full JFPD beats standard fine-tuning in every setting; on the CNN, gains to Syn are +19.25 (MNIST), +12.35 (SVHN) and +9.55 (USPS) points.
- Prediction-guided feature alignment alone (PGFD) is unstable, especially on ViT-S: both views are needed.

Comparison with state of the art (ViT-B/16 backbone)

TABLE II: Comparison with SOTA methods on Office-Home.

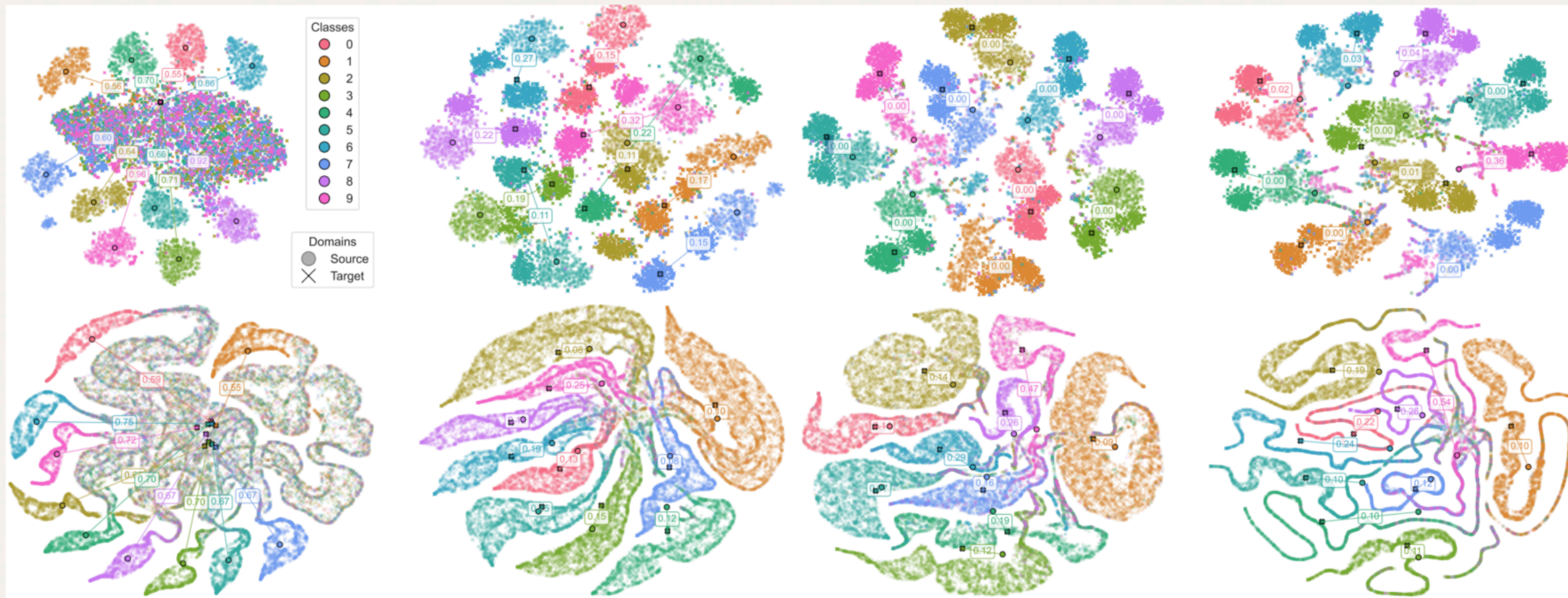
Method	Ar2Cl	Ar2Pr	Ar2Re	Cl2Ar	Cl2Pr	Cl2Re	Pr2Ar	Pr2Cl	Pr2Re	Re2Ar	Re2Cl	Re2Pr	Avg.
ResNet-50 [39]	44.9	66.3	74.3	51.8	61.9	63.6	52.4	39.1	71.2	63.8	45.9	77.2	59.4
MinEnt [40]	51.0	71.9	77.1	61.2	69.1	70.1	59.3	48.7	77.0	70.4	53.0	81.0	65.8
SAFN [41]	52.0	71.7	76.3	64.2	69.9	71.9	63.7	51.4	77.1	70.9	57.1	81.5	67.3
CDAN+E [24]	54.6	74.1	78.1	63.0	72.2	74.1	61.6	52.3	79.1	72.3	57.3	82.8	68.5
DCAN [42]	54.5	75.7	81.2	67.4	74.0	76.3	67.4	52.7	80.6	74.1	59.1	83.5	70.5
BNM [43]	56.7	77.5	81.0	67.3	76.3	77.1	65.3	55.1	82.0	73.6	57.0	84.3	71.1
SHOT [44]	57.1	78.1	81.5	68.0	78.2	78.1	67.4	54.9	82.2	73.3	58.8	84.3	71.8
ATDOC-NA [45]	58.3	78.8	82.3	69.4	78.2	78.2	67.1	56.0	82.7	72.0	58.2	85.5	72.2
CoWA-JMDS [37]	56.9	78.4	81.0	69.1	80.0	79.9	67.7	57.2	82.4	72.8	60.5	84.5	72.5
Fixbi [33]	58.1	77.3	80.4	67.7	79.5	78.1	65.8	57.9	81.7	76.4	62.9	86.7	72.7
DUET [46]	73.6	90.4	91.0	83.6	90.7	90.9	82.7	73.7	91.2	83.6	74.0	91.2	84.7
ViT-B [47]	54.7	83.0	87.2	77.3	83.4	85.6	74.4	50.9	87.2	79.6	54.8	88.8	75.5
CDTrans-B [48]	68.8	85.0	86.9	81.5	87.1	87.3	79.6	63.3	88.2	82.0	66.0	90.6	80.5
TVT-B [49]	74.9	86.8	89.5	82.8	88.0	88.3	79.8	71.9	90.1	85.5	74.6	90.6	83.6
SSRT-B [50]	75.2	89.0	91.1	85.1	88.3	90.0	85.0	74.2	91.3	85.7	78.6	91.8	85.4
DCST [51]	76.1	90.6	91.3	86.5	91.4	92.3	84.5	73.3	91.7	86.0	74.4	93.2	86.0
TSGF-B [52]	81.2	92.6	92.3	86.4	92.7	92.1	84.4	79.8	93.3	87.2	82.1	94.2	88.2
FFTAT-B [53]	83.2	92.9	95.2	91.1	93.5	95.2	89.7	85.0	94.9	93.0	87.5	95.9	91.4
JFPD (ours)	86.9	92.9	95.4	91.4	94.0	95.6	90.2	86.6	95.2	93.1	88.4	96.5	92.2

TABLE III: Comparison with SOTA methods on VisDA-17.

Method	plane	bcycl	bus	car	horse	knife	mcycl	person	plant	sktbrd	train	truck	Avg.
DANN [29]	81.9	77.7	82.8	44.3	81.2	29.5	65.1	28.6	51.9	54.6	82.8	7.8	57.4
MinEnt [40]	80.3	75.5	75.8	48.3	77.9	27.3	69.7	40.2	46.5	46.6	79.3	16.0	57.0
BNM [43]	89.6	61.5	76.9	55.0	89.3	69.1	81.3	65.5	90.0	47.3	89.1	30.1	70.4
CDAN+E [24]	85.2	66.9	83.0	50.8	84.2	74.9	88.1	74.5	83.4	76.0	81.9	38.0	73.9
SAFN [41]	93.6	61.3	84.1	70.6	94.1	79.0	91.8	79.6	89.9	55.6	89.0	24.4	76.1
CGDM [54]	93.7	82.7	73.2	68.4	92.9	94.5	88.7	82.1	93.4	82.5	86.8	49.2	82.3
SHOT [44]	94.3	88.5	80.1	57.3	93.1	93.1	80.7	80.3	91.5	89.1	86.3	58.2	82.9
CoWA-JMDS [37]	96.2	89.7	83.9	73.8	96.4	97.4	89.3	86.8	94.6	92.1	88.7	53.8	86.9
FixBi [33]	96.1	87.8	90.5	90.3	96.8	95.3	92.8	88.7	97.2	94.2	90.9	25.7	87.2
CAN [55]	97.0	87.2	82.5	74.3	97.8	96.2	90.8	80.7	96.6	96.3	87.5	59.9	87.2
TVT-B [49]	92.9	85.6	77.5	60.5	93.6	98.2	89.4	76.4	93.6	92.0	91.7	55.7	83.9
CDTrans-B [48]	97.1	90.5	82.4	77.5	96.6	96.1	93.6	88.6	97.9	86.9	90.3	62.8	88.4
SSRT-B [50]	98.9	87.6	89.1	84.8	98.3	98.7	96.3	81.1	94.9	97.9	94.5	43.1	88.8
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FFTAT-B [53]	99.7	98.5	93.1	81.1	99.8	99.5	97.8	89.6	95.7	99.8	98.7	72.4	93.8
JFPD (ours)	100.0	98.8	93.0	76.2	99.9	99.5	97.8	88.5	97.8	99.9	99.2	80.9	94.3

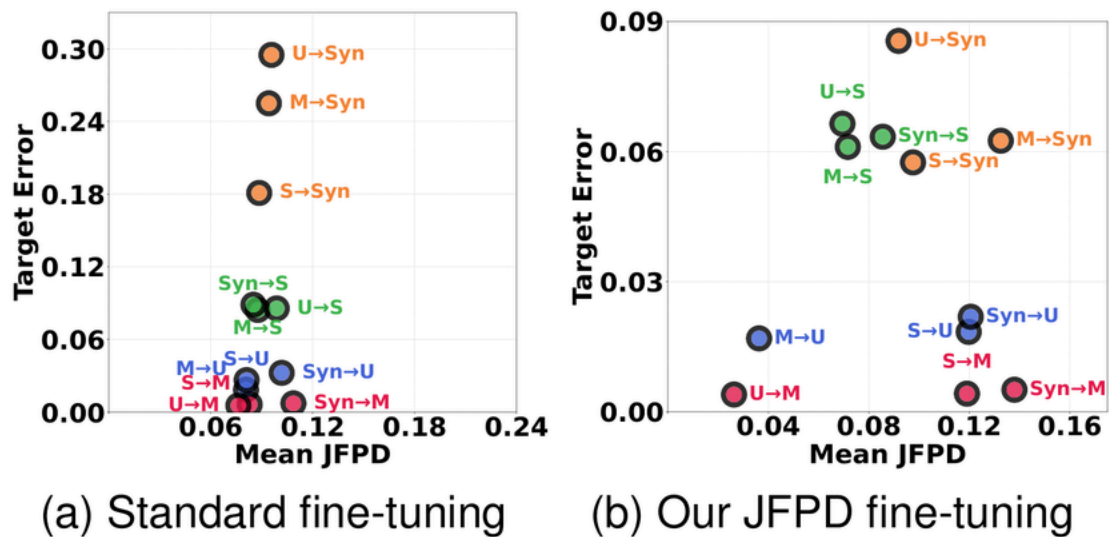
Office-Home 92.2% (FFTAT-B 91.4%) · VisDA-2017 94.3% (FFTAT-B 93.8%) · DomainNet 52.3% (DSAN-B 52.0%)

Outcome: The model learns to focus on reliable, trustworthy samples, effectively bridging the domain gap

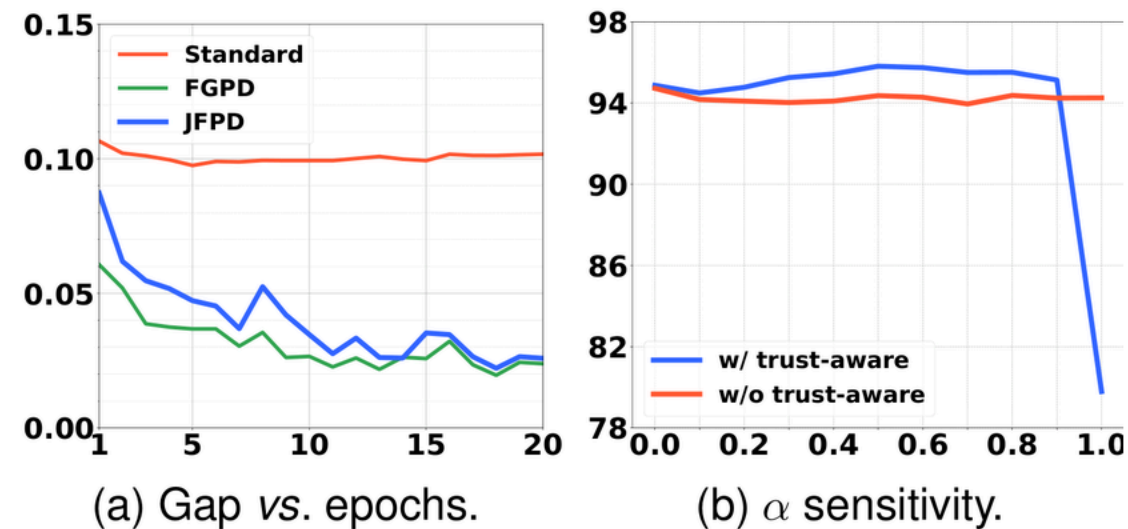


t-SNE visualizations of MNIST (source, dots) and SVHN (target, crosses). The first row shows learned feature representations; the second row shows softmax predictions. From left to right: (i) pretrained MNIST model (no SVHN fine-tuning), (ii) standard fine-tuning, (iii) our feature-guided prediction divergence, and (iv) our full Joint Feature-Prediction Discrepancy (JFPD) method.

JFPD as a diagnostic, and why trust matters



Mean JFPD vs. target error on Digits (each point = one source→target transfer). Larger JFPD means harder adaptation, and the relation becomes much cleaner after JFPD training.



(a) JFPD reduces the domain gap fastest and lowest over epochs.
(b) Accuracy is stable for $\alpha \in [0.1, 0.9]$ (best $\approx 95.8\%$ at $\alpha \approx 0.5-0.6$) and consistently above the version without trust weights.

Synthesis & The Big Picture

JFPD creates harmony between features and predictions, guided by trust





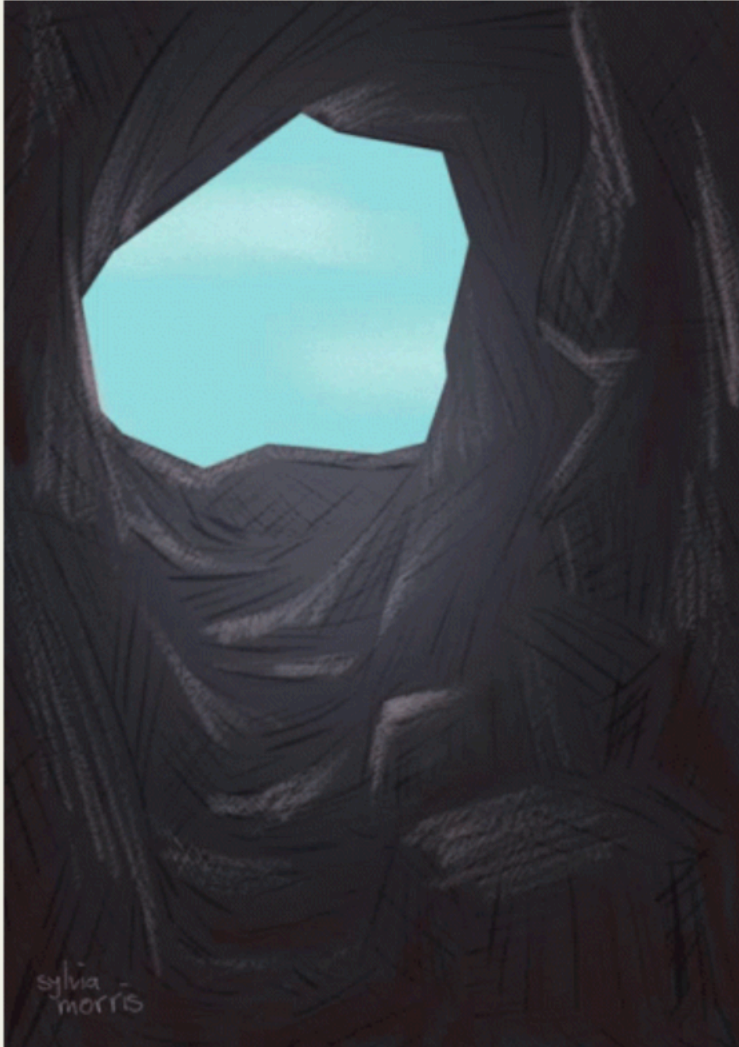
Key Takeaways

JFPD uses trust to selectively align features and predictions in new domains.

Its trust-weighted discrepancy tracks target error, making it a diagnostic for adaptation difficulty.

State-of-the-art accuracy on Digits, Office-Home, VisDA-2017 and DomainNet.

The Echo Effects



- **More Robust:** Better at handling real-world data shifts.
- **More Generalizable:** Perform well even with limited labels.
- **More Interpretable:** The feature space becomes more semantically meaningful.

Aligning what a model sees with what it believes is a fundamental step toward more human-like learning

Thanks for watching!

